

BASANT BHATTARAI

Senior Data & AI Engineer | Data Platform & DBA

Address: Kathmandu, Nepal

Email: technobasant9@gmail.com

LinkedIn: linkedin.com/in/technobasant

GitHub: github.com/technobasant

SUMMARY

Senior data & AI engineer with **9+ years** building and operating scalable, low-latency data platforms processing **15TB+ and hundreds of millions of events daily** across AWS, GCP, and Azure. **Open-source-first** — Kafka, Spark, Apache Iceberg, Trino, dbt, and Airflow on Kubernetes — across **ingestion, transformation, and BI/reverse-ETL consumption**, with the **governance, observability, and reliability (SLOs, 99.9% uptime)** production data products demand. Also delivers production **AI/agent systems** (LangChain, CrewAI, Google-ADK, RAG) and administers the **databases** behind them — PostgreSQL, MongoDB, Cassandra/ScyllaDB, Redis, MySQL, ClickHouse (HA, backup/DR, tuning, security). Track record: **40% lower cost, 50% faster processing, 60% faster analytics delivery.**

TECHNICAL SKILLS

Data Engineering: Batch & stream ingestion, transformation and consumption layers, forward/reverse-ETL, ETL/ELT, Data Lakehouse, Lambda Architecture, Data Mesh, data modeling, data contracts, schema evolution, SCD Types 1/2/3, data quality, data lineage, data governance, data cataloging

Big Data & Processing: Apache Spark, PySpark, Databricks, Delta Lake, Apache Iceberg, Hadoop, Hive, dbt, stream/batch processing; object storage (S3/GCS/ADLS)

AI, GenAI & Agent Systems: Google-ADK, LangChain, LangGraph, CrewAI, LlamaIndex, AutoGen, Haystack, OpenAI Agents SDK, Semantic Kernel, DSPy, RAG, LLM Fine-Tuning, MCP, ReAct, Prompt/Context Engineering, AI Governance, Guardrails; Vector DBs (Milvus, Pinecone, pgvector, Chroma, FAISS, Weaviate)

Streaming & Orchestration: Apache Kafka, Apache Flink, Kinesis, Apache Beam, Apache Airflow / Cloud Composer, Prefect, CDC (Debezium/Datastream)

Data Platform & Reliability Ops: SLOs/SLAs, freshness & latency monitoring, capacity planning, cost/FinOps optimization, backup/DR & RPO/RTO, incident management, runbooks, change management, on-call

Observability: Prometheus, Grafana, Datadog, OpenTelemetry (pipeline freshness, lag, throughput, data-quality SLOs); LangSmith, LangFuse, Opik (AI/agent tracing); MLflow

Databases & DBA: PostgreSQL, MongoDB, Cassandra/ScyllaDB, Redis, MySQL/MariaDB, ClickHouse, Snowflake, BigQuery, Redshift, Trino (PrestoSQL), DuckDB, Citus, TimescaleDB, Elasticsearch, Druid; HA & replication/clustering, sharding, backup/DR & PITR, performance tuning & indexing, RBAC/security & auditing, version upgrades

Security & Governance: RBAC, authentication/authorization, encryption at rest & in transit, access auditing/logging, least-privilege reviews, secrets management, Unity Catalog, GDPR-aware data governance

Cloud Platforms:

AWS: S3, Lambda, EKS, Bedrock, SageMaker, Glue, Redshift, Athena, Kinesis, EMR, Step Functions

GCP: BigQuery, Dataflow, Dataproc, Datastream, Vertex AI, Pub/Sub, Cloud Composer, GKE

Azure: Synapse Analytics, Data Factory, Databricks, ADLS, Event Hubs, AKS, Azure ML, Power BI

Infrastructure & DevOps: Kubernetes, Docker, Helm, Terraform, Ansible, Jenkins, GitLab CI/CD, GitHub Actions, Infrastructure as Code, DataOps, Microservices, Serverless

Programming: Python, SQL, Scala, Java, Bash, JavaScript

Libraries & Frameworks: FastAPI, FastMCP, Flask, Django, Node.js, Express, Next.js, React, PySpark, Pandas, NumPy, Polars, PyArrow, Pydantic, SQLAlchemy, Celery, PyTorch

PROFESSIONAL EXPERIENCE

Senior Data Engineer | UXCam | Kathmandu, Nepal (Global Team) | February 2024 - Present

- Architected and operate the company data platform on **self-managed Spark clusters on Kubernetes** handling **10TB+ and hundreds of millions of events daily** — Kafka **ingestion**, Spark/PySpark + **dbt/Airflow transformation**, and BI/**reverse-ETL consumption** serving **25K+ mobile apps** and stakeholders
- Run **Databricks + Unity Catalog** (AWS/GCP) as a managed complement for collaborative analytics, governed data sharing, and ML workflows
- Established **data platform reliability & observability** — freshness/latency **SLOs**, data-quality checks, and lineage via Prometheus/Grafana/OpenTelemetry; sustain **99.9% uptime** with **RBAC, encryption in transit/at rest, and GDPR-aware governance**
- Built and own the **App Analytics Agent Platform** (Google-ADK, LangChain, CrewAI, MCP, LangGraph orchestration) across **25K+ apps**, cutting analytics delivery time **60%** and analyst manual effort **75%**
- Built **RAG pipelines on Milvus** processing **1M+ videos/month** (**20%** higher engagement insights) and drove **MLOps** with MLflow (recommendation personalization **+25%**)
- Mentor **5 engineers** through code reviews and cross-team pairing; team delivery times decreased **30%**

Data Engineer | UXCam | Kathmandu, Nepal | February 2020 - February 2024

- Rebuilt the core **ETL layer** in Spark/PySpark from the ground up and automated orchestration with **Airflow**; processing time for **5TB+/day** of mobile analytics data dropped **50%**, unblocking real-time product dashboards
- Designed the **Iceberg-based Lakehouse** with SCD patterns, now handling **petabyte-scale** data at **35%** lower storage cost
- Tuned distributed query engines and databases — **Trino, DuckDB, Citus, TimescaleDB, ClickHouse** — serving **100M+ queries/day**; query optimization and repartitioning delivered **50%** speed-up
- Built **Kinesis Data Streams** ingestion with **Spark Structured Streaming**, with a **data-quality framework holding 99.5% accuracy**; handled schema evolution and CDC into downstream stores
- Led the **Kubernetes migration** of the full data infrastructure: **40% cost reduction, 99.9% uptime**, and auto-scaling that held under peak load
- Operated production workloads on **AWS** (Lambda, EC2, RDS, EKS, S3, Redshift, Glue, Athena) with monitoring and alerting; collaborated with US/EU stakeholders across timezones

Project Leader | SVCET | India | 2019 - 2020

- Led a team of 4 developers on an Intelligent Dialogue System; NLP + deep learning achieved **90% intent recognition**, meeting the client acceptance threshold
- Coordinated requirements, sprint planning, and delivery milestones across academic and industry stakeholders; project completed **two weeks ahead of schedule**

Software Developer (Backend & Data) | SV Technology | India | 2017 - 2019

- Built backend APIs in **Python/FastAPI** serving 10K+ users at **85% test coverage**, plus early data-pipeline work that shaped the move into data engineering
- Designed and maintained **PostgreSQL** schemas and stored procedures for transactional and reporting systems; contributed to **CI/CD** setup with Jenkins

EDUCATION

Bachelor of Technology in Computer Science and Engineering

JNTUA College of Engineering, Anantapur, India | 2015 - 2019

REFERENCES

Available upon request.